

Digital Logic Circuits (Spring 2026)

HW #4 (Due: May 6, 2026)

8. Two edge-triggered J-K flip-flops are shown in Figure 7-77. If the inputs are as shown, draw the Q output of each flip-flop relative to the clock, and explain the difference between the two. The flip-flops are initially RESET.

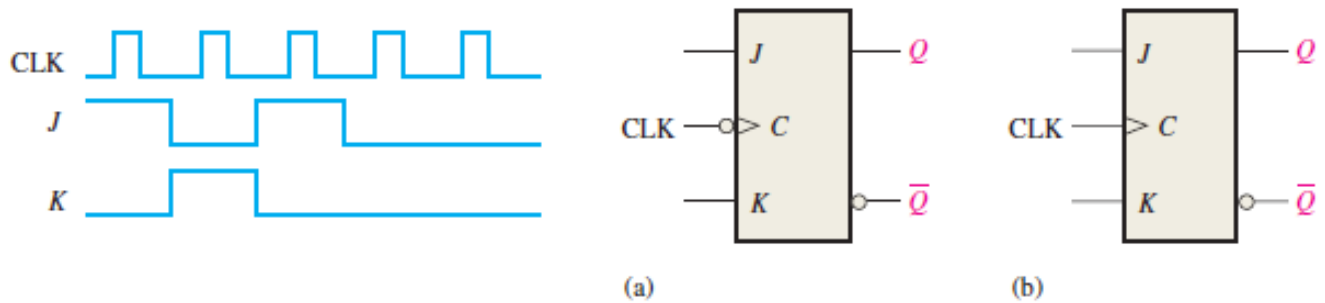


FIGURE 7-77

10. Draw the Q output relative to the clock for a D flip-flop with the inputs as shown in Figure 7-79. Assume positive edge-triggering and Q initially LOW.

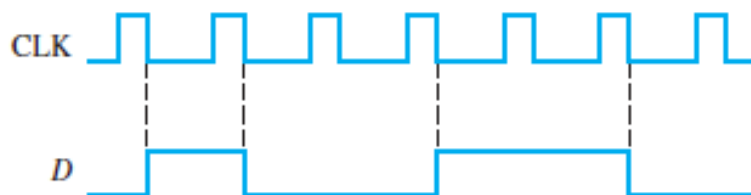


FIGURE 7-79

14. Determine the Q waveform relative to the clock if the signals shown in Figure 7–83 are applied to the inputs of the J-K flip-flop. Assume that Q is initially LOW.

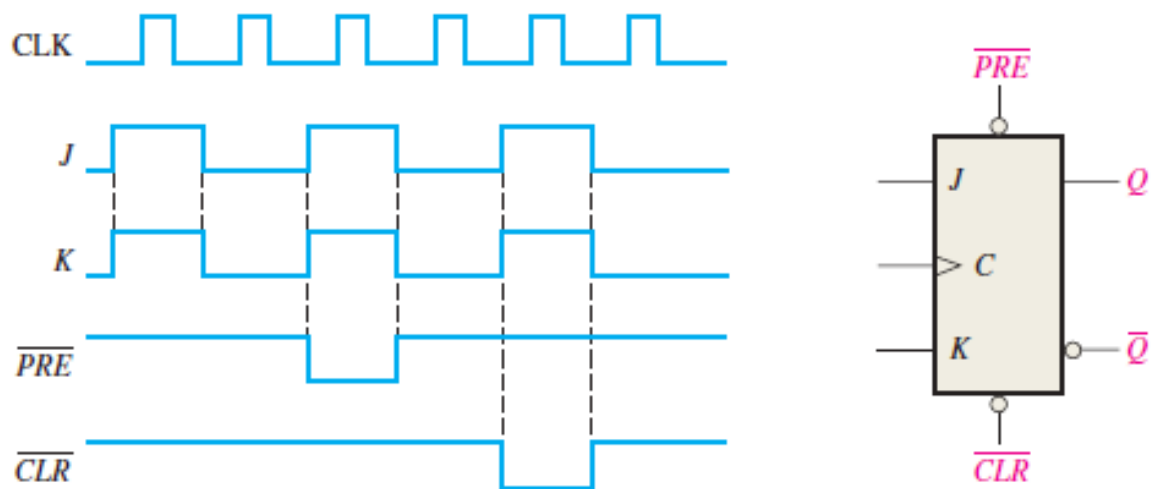


FIGURE 7–83

16. The following serial data are applied to the flip-flop through the AND gates as indicated in Figure 7–85. Determine the resulting serial data that appear on the Q output. There is one clock pulse for each bit time. Assume that Q is initially 0 and that $\overline{PRE}$ and $\overline{CLR}$ are HIGH. Right-most bits are applied first.

J_1 : 1 0 1 0 0 1 1; J_2 : 0 1 1 1 0 1 0; J_3 : 1 1 1 1 0 0 0; K_1 : 0 0 0 1 1 1 0; K_2 : 1 1 0 1 1 0 0;
 K_3 : 1 0 1 0 1 0 1

17. For the circuit in Figure 7–85, complete the timing diagram in Figure 7–86 by showing the Q output (which is initially LOW). Assume $\overline{PRE}$ and $\overline{CLR}$ remain HIGH.

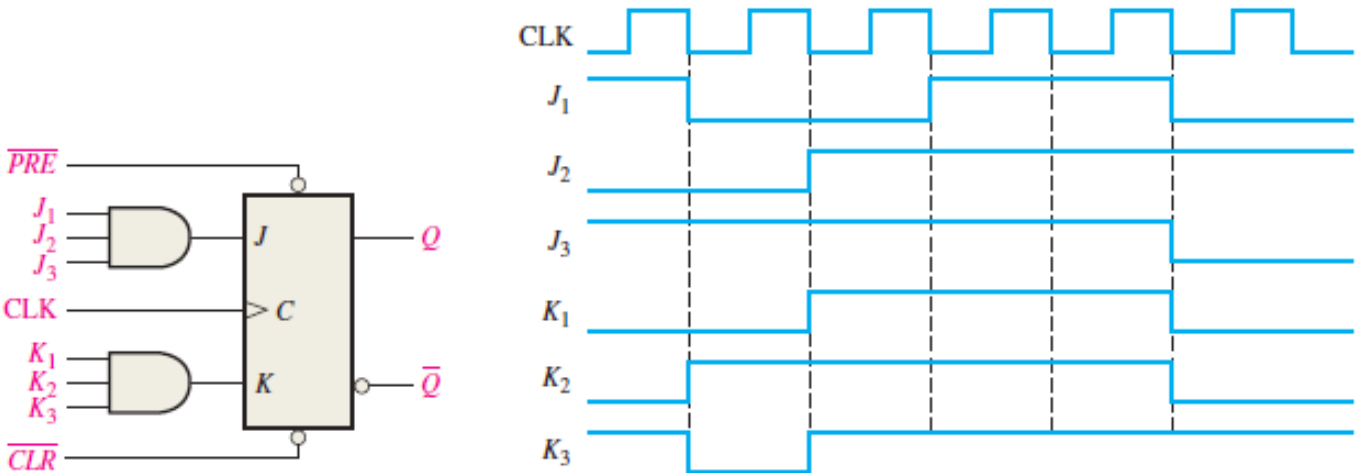


FIGURE 7–85

FIGURE 7–86

25. A D flip-flop is connected as shown in Figure 7-90. Determine the Q output in relation to the clock. What specific function does this device perform?

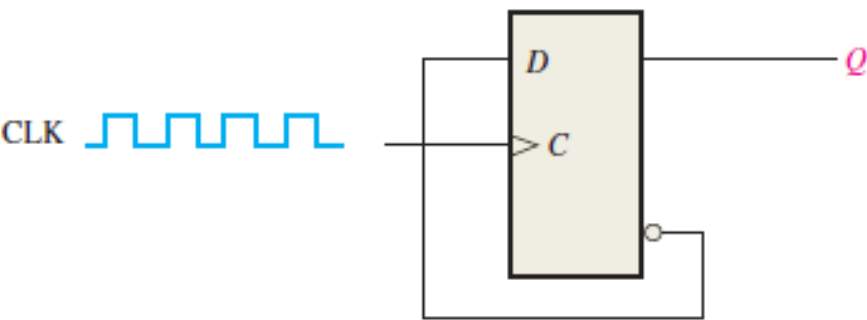


FIGURE 7-90

7. What is the state of the register in Figure 8-49 after each clock pulse if it starts in the 101001111000 state?

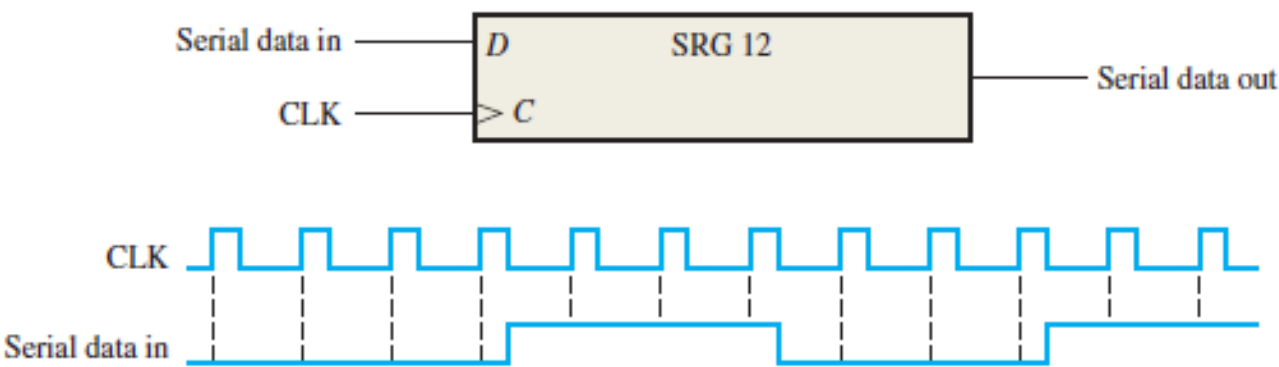


FIGURE 8-49

10. A leading-edge clocked serial in/serial out shift register has a data-output waveform as shown in Figure 8-52. What binary number is stored in the 8-bit register if the first data bit out (left-most) is the LSB?

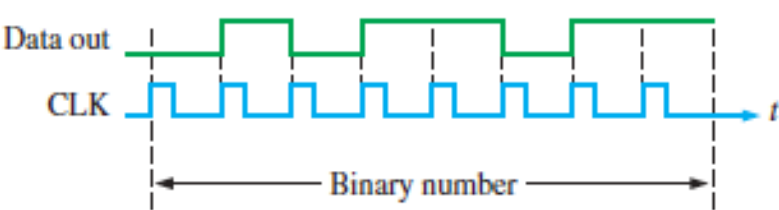


FIGURE 8-52

28. The waveform pattern in Figure 8–61 is required. Devise a ring counter, and indicate how it can be preset to produce this waveform on its Q_9 output. At CLK16 the pattern begins to repeat.

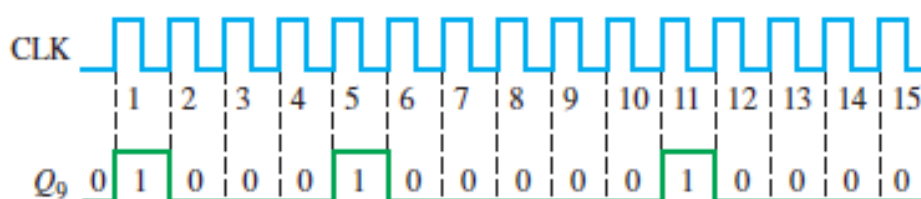
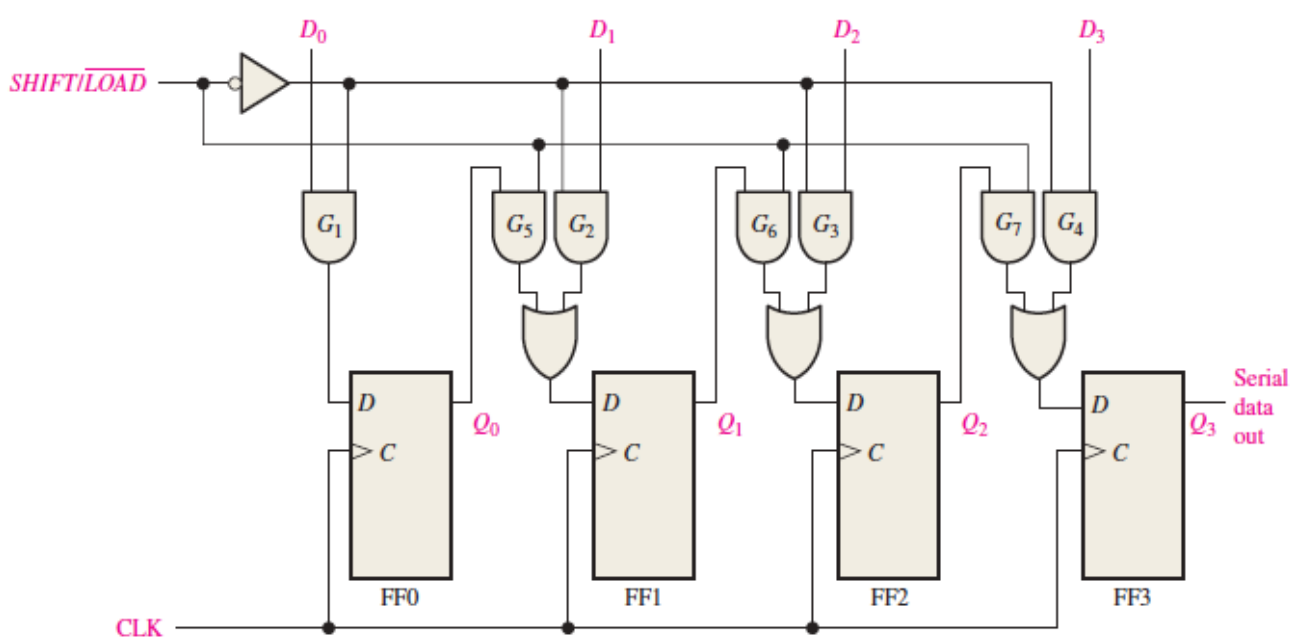
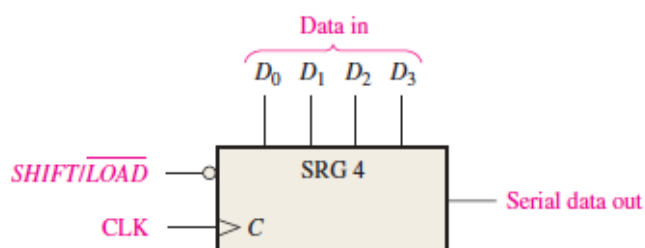


FIGURE 8–61



(a) Logic diagram



(b) Logic symbol

FIGURE 8–10 A 4-bit parallel in/serial out shift register. Open file F08-10 to verify operation.



33. Refer to the parallel in/serial out shift register in Figure 8–10. The register is in the state where $Q_0Q_1Q_2Q_3 = 1001$, and $D_0D_1D_2D_3 = 1010$ is loaded in. When the $\overline{SHIFT/LOAD}$ input is taken HIGH, the data shown in Figure 8–63 are shifted out. Is this operation correct? If not, what is the most likely problem?

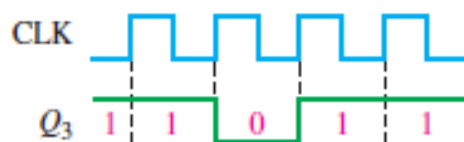


FIGURE 8–63

8. Two edge-triggered J-K flip-flops are shown in Figure 7-77. If the inputs are as shown, draw the Q output of each flip-flop relative to the clock, and explain the difference between the two. The flip-flops are initially RESET.

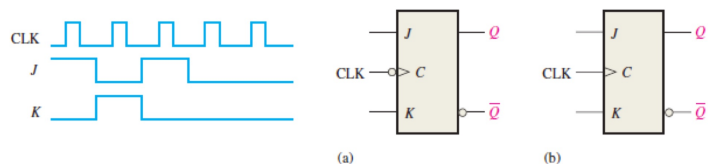


FIGURE 7-77

10. Draw the Q output relative to the clock for a D flip-flop with the inputs as shown in Figure 7-79. Assume positive edge-triggering and Q initially LOW.

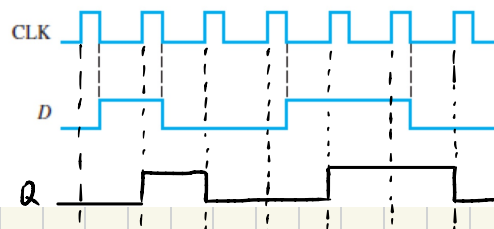


FIGURE 7-79

14. Determine the Q waveform relative to the clock if the signals shown in Figure 7-83 are applied to the inputs of the J-K flip-flop. Assume that Q is initially LOW.

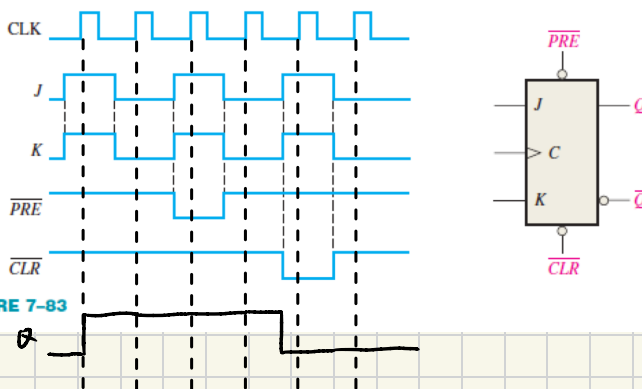


FIGURE 7-83

7. What is the state of the register in Figure 8-49 after each clock pulse if it starts in the 1010111000 state?

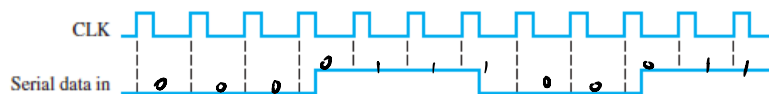
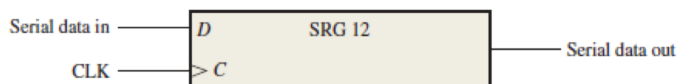


FIGURE 8-49

0	1	0	1	0	0	1	1	1	0	0	0
1	0	1	0	1	0	0	1	1	1	0	0
2	0	0	1	0	1	0	0	1	1	1	0
3	0	0	0	1	0	1	0	0	1	1	1
4	0	0	0	0	1	0	1	0	0	1	1
5	1	0	0	0	0	1	0	1	0	0	1
6	1	1	0	0	0	0	1	0	1	0	0
7	1	1	1	0	0	0	0	1	0	1	0
8	0	1	1	1	0	0	0	0	1	0	1
9	0	0	1	1	1	0	0	0	0	1	1
10	0	0	0	1	1	1	0	0	0	0	1
11	1	0	0	0	1	1	1	0	0	0	1
12	1	1	0	0	0	1	1	1	0	0	0

10. A leading-edge clocked serial in/serial out shift register has a data-output waveform as shown in Figure 8-52. What binary number is stored in the 8-bit register if the first data bit out (left-most) is the LSB?

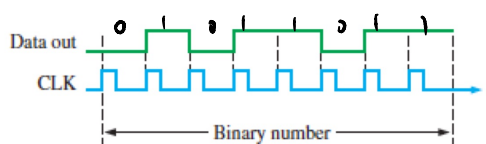


FIGURE 8-52

Ans: 11011010

28. The waveform pattern in Figure 8-61 is required. Devise a ring counter, and indicate how it can be preset to produce this waveform on its Q_9 output. At CLK 16 the pattern begins to repeat.

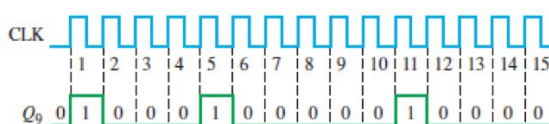


FIGURE 8-61

Q₁ 0001000010000100
0001000010000100
01234567891011121314

33. Refer to the parallel in/serial out shift register in Figure 8-10. The register is in the state where $Q_0Q_1Q_2Q_3 = 1001$, and $D_0D_1D_2D_3 = 1010$ is loaded in. When the $\overline{SHIFT/LOAD}$ input is taken HIGH, the data shown in Figure 8-63 are shifted out. Is this operation correct? If not, what is the most likely problem?

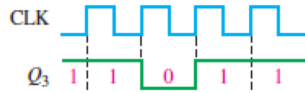


FIGURE 8-63

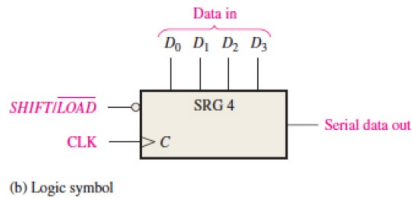
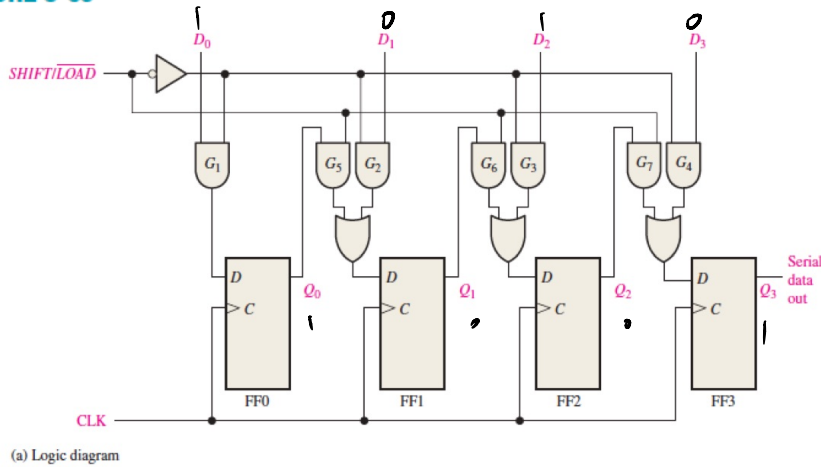


FIGURE 8-10 A 4-bit parallel in/serial out shift register. Open file F08-10 to verify operation.



Not correct.

When $\overline{SHIFT/LOAD}$ is Low

D_1, D_2, D_3, D_4 is loaded into register

$Q_0Q_1Q_2Q_3 = 1010$

Then $\overline{SHIFT/LOAD}$ is High with CLK

$Q_3 = 0 \rightarrow Q_3 = Q_2 = 1 \rightarrow Q_3 = Q_1 = 0$

$\rightarrow Q_3 = Q_0 = 1$

Should be: $0 \rightarrow 1 \rightarrow 0 \rightarrow 1$

But in real: $1 \rightarrow 1 \rightarrow 0 \rightarrow 1$

Most Likely problem = D_3 was not correctly loaded into FF3.